<u>The tasks for lab report 3:</u> Normal Structure of the lab report (same for all the lab reports):

SL No.	Sections	What to Write
1.	Cover Page	Add a cover page and use the format uploaded in the drive (Lab Report Format (with cover page).docx)
2.	Objective	Write 2-3 lines using points to mention what was the main learning outcome of the report
3.	Components	Write what types of components you used.
4.	Theory	Write some theoretical background if needed for the problems you are going to solve in this report
5.	Problems mention in the tasks	Solve the following questions with proper answers
6.	Discussion	Write 5-7 lines on what you learn and what problems you faced when solving these problems.

Please Answer the following questions for the Lab Report Task 4:

1) Write a python code in the 'thonny' editor that can access the RPi camera and stream live video. Also, write another code that can use your mobile camera as a webcam to stream live video.
   In both cases, if 'q' is pressed, the streaming will be stopped.
2) Say you want to run a customized machine learning problem where you build a low cost security system for the software firm you are working. Your system should be able to identify the employees of your firm and can successfully detect people outside your firm. You have built your datasets and trained your ML Model on a local computer which gave very good accuracy. Now, Write in **details** and explain the step by step process on how this **Customized** machine learning model can be run on ESP32 while using ESP camera. For your reference, As ESP32 is a very small scale processor, TinML can be run on ESP32. (TinyML: Implementing ML Models on ESP32 | by Manjit Baishya | Medium)
   You have to write in as many details as you can provide while searching google and other internet sources in such a way that <u>anyone can run a simple ML model using ESP32 easily by reading your explained processes provided in this lab report.</u>

   You can use step by step points. Give a reference Link for each process you explained in this part.

3) You saw the objection detection experiment at the end of the Experiment 04. You saw that there were a total of 5 steps to complete the overall process. In step 2, you download the code from the github repository and in step 5, you download the ML model. Explain in detail why you have to download the ML model in step 5 while you have already downloaded the codes in step 2.

4) In the object detection part in experiment 4, you have done the followings:

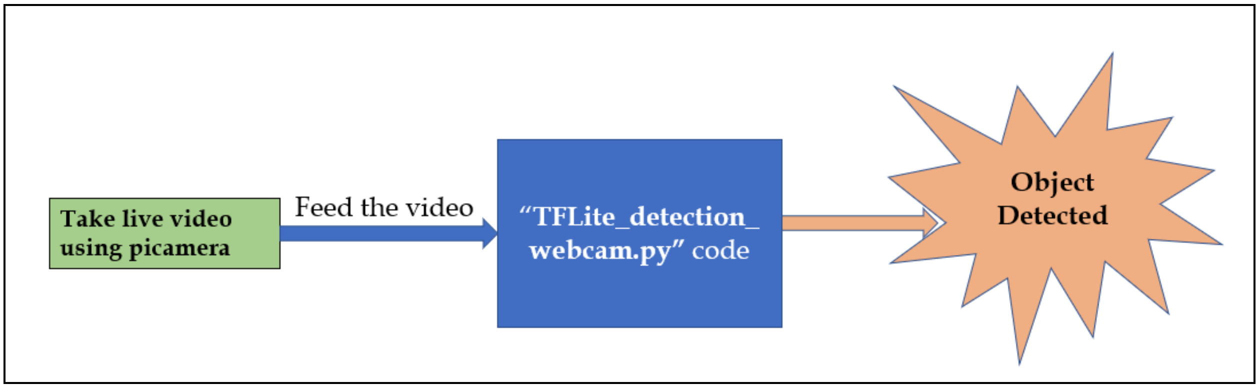

Fig. 3 for problem no. 3

Now, Explain in detail how the live video is being processed inside the "TFLite_detection_webcam.py" code and classify/detect objects.

[N:B: Please note that problems mentioned here are fully theoretical and coding related. So, you have to search the internet and other verified sources, mention/cite these sources properly. Also, explain the whole process in each problem in a clear manner. Do not just write unrelated and useless things which are unnecessary.]